

ANDY UNG

✉ ung.andy01@gmail.com ·  Andy Ung ·  andyung17

EDUCATION

Georgia Institute of Technology

Online Masters of Computer Science, Artificial Intelligence Specialization

Ottawa, ON

Jan 2026 - Present

University of Ottawa GPA 3.9/4.0 7x Dean's Honours List

Bachelor of Applied Science, Software Engineering

Ottawa, ON

Sept 2019 - July 2024

RELEVANT EXPERIENCE

Middleware Software Developer

Ford Motor Company

Ottawa, ON

May 2023 - Sep 2023

- Engineered a high-throughput data pipeline using Python and RabbitMQ to stream ROS telemetry across **50+** vehicle hardware features and **1000+** lines of telemetry data, facilitating high-fidelity simulation testing
- Architected a C++ API to collect **52+** real-time metrics from critical vehicle hardware, enabling live diagnostic overlays on the vehicle display
- Authored test cases and test plans for a vehicle simulation suite using pytest, increasing test coverage by **30+** and surfacing critical defects before deployment, improving vehicle safety and reliability

Backend Software Developer

Bluwave-AI

Ottawa, ON

May 2022 - Sep 2022

- Built a data ingestion pipeline using Python, Node.js, and OpenCV to digitize legacy forecast charts, establishing a second validated data source used to cross-check and supplement the primary training dataset
- Optimized inference logic to dynamically bypass low-value nighttime predictions, reducing cloud computational cost by **25%** with zero impact on model accuracy
- Monitored Kubernetes pod health via Prometheus, Grafana, and CLI tooling to maintain observability across a distributed microservice architecture

Full-Stack Software Developer

Government of Canada | Transport Canada

Ottawa, ON

Sept 2021 - Apr 2022

- Optimized a relational database (Microsoft SQL Server), reducing average query execution time from **900ms to 300ms** through query refactoring and index tuning
- Shipped **5** core feature updates to an aircraft information system on Azure, gathering requirements through stakeholder meetings to streamline internal reporting workflows
- Architected a data access layer using the DAO pattern in ASP.NET, decoupling business logic from data persistence to improve modularity, simplify unit testing, and eliminate duplicate query code

Software Designer

Arkalumen

Ottawa, ON

Jan 2021 - Apr 2021

- Designed an extensible hardware abstraction layer in C#/MVVM Light Toolkit, enabling support for **10+** LED controller hardware models without requiring core product rewrites
- Built real-time synchronization middleware between Firebase and a desktop application, syncing configuration data within seconds across **50+** customer deployments

SIDE PROJECTS

SteamPA Vue.js, Python, Java, TypeScript, Steam OAuth API

- Building a full-stack web dashboard that authenticates users via Steam OAuth and visualizes recently played games and top genres
- Designing backend data processing in Python and Java to parse Steam API data and aggregate genre/playtime statistics

TECHNICAL SKILLS

- **Languages:** Python, C++, C#, Java, Golang, JavaScript, Kotlin, HTML, CSS
- **Frameworks & Tools:** React, Angular, Node.js, ASP.NET, Docker, Kubernetes, GCP, Jenkins
- **Databases:** Microsoft SQL Server, relational databases